

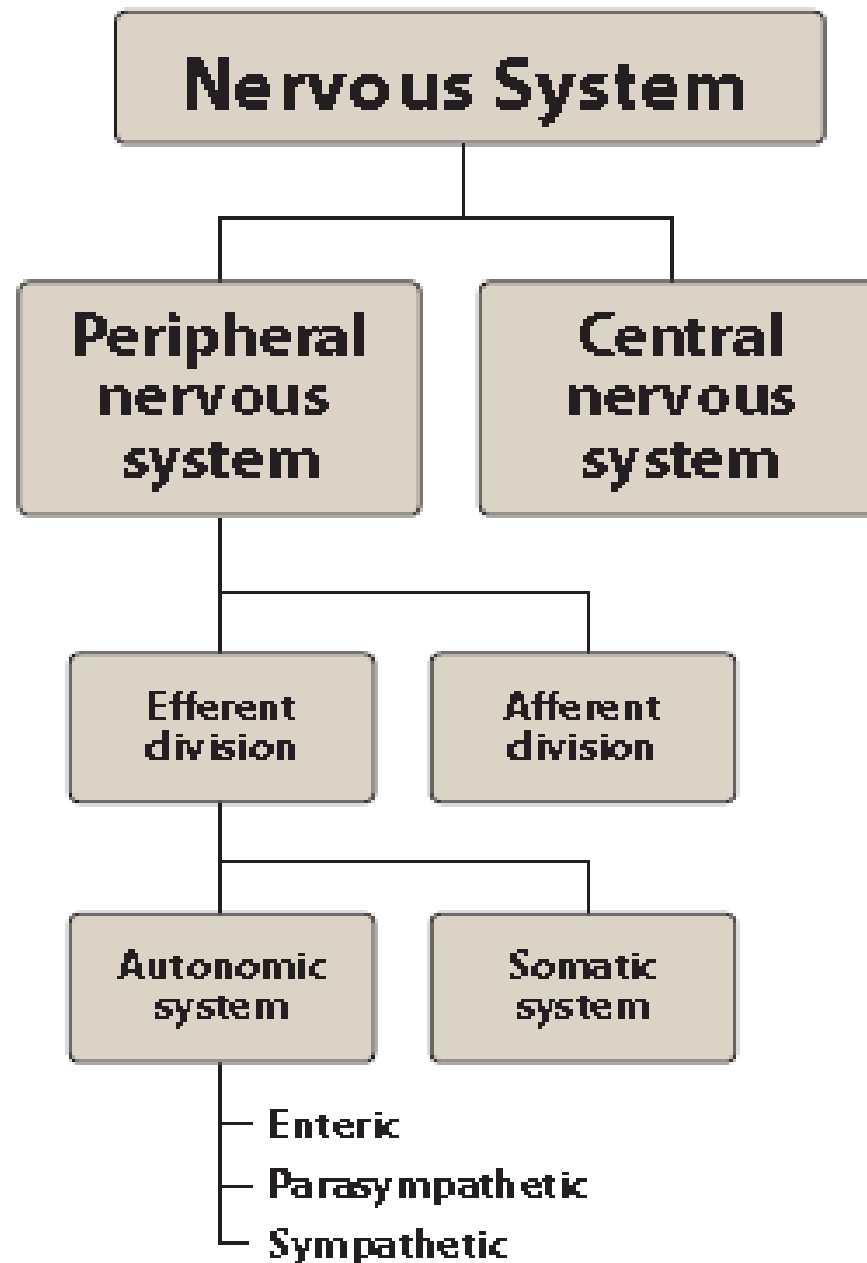
DRUGS AFFECTING THE AUTONOMIC NERVES SYSTEM

By

Yaschilal Muche (B.Pharm, MSc)

Assistant Professor of Pharmacology, WU

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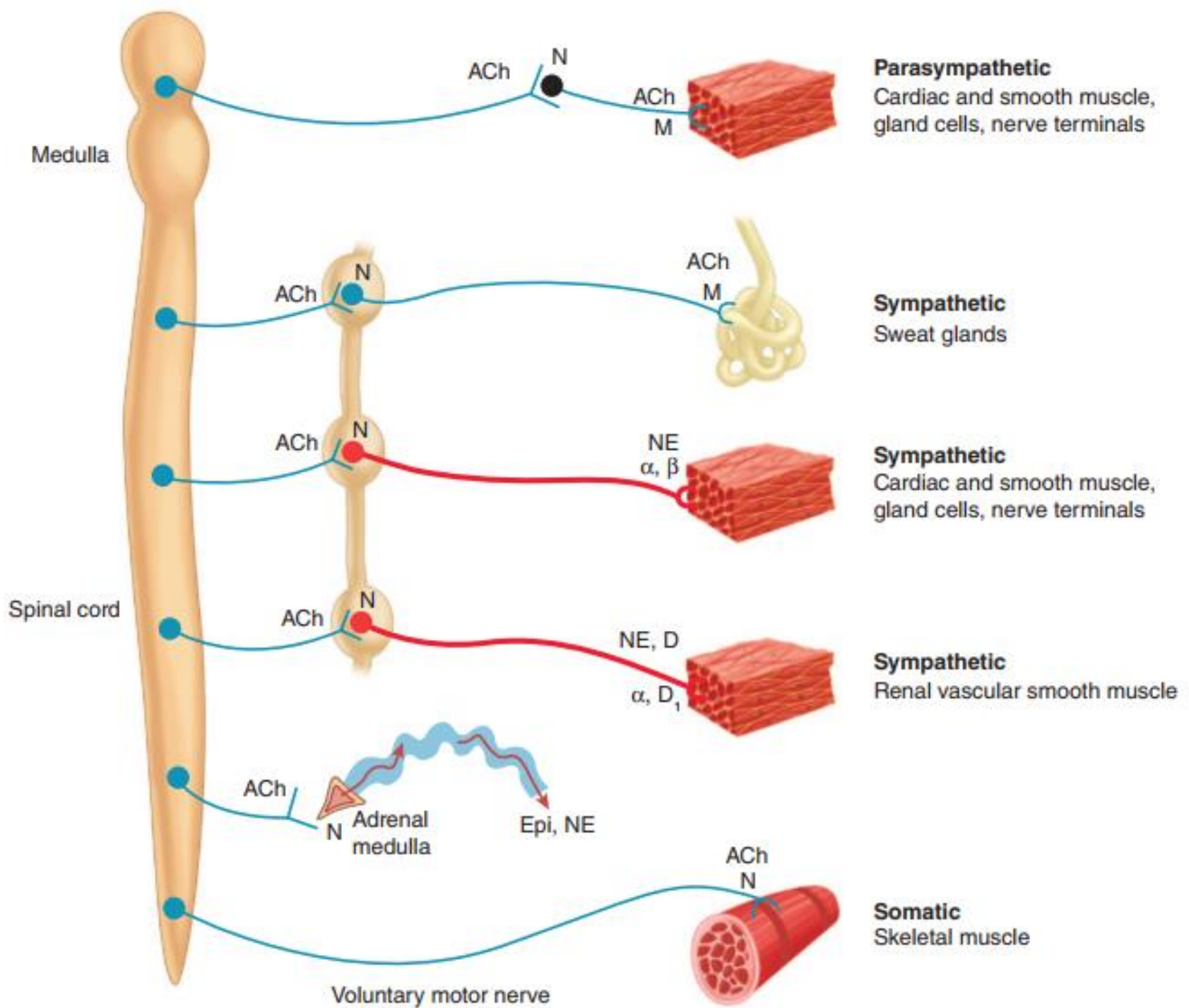
Autonomic Nervous System

❑ Sympathetic (adrenergic)

- Exit from thoracic and lumbar regions
- Neurotransmitter- norepinephrine (predominant)
- Preganglionic neuron is shorter than postganglionic neuron
- Adrenergic receptors- **$\alpha 1$, $\alpha 2$, $\beta 1$, $\beta 2$, $\beta 3$**

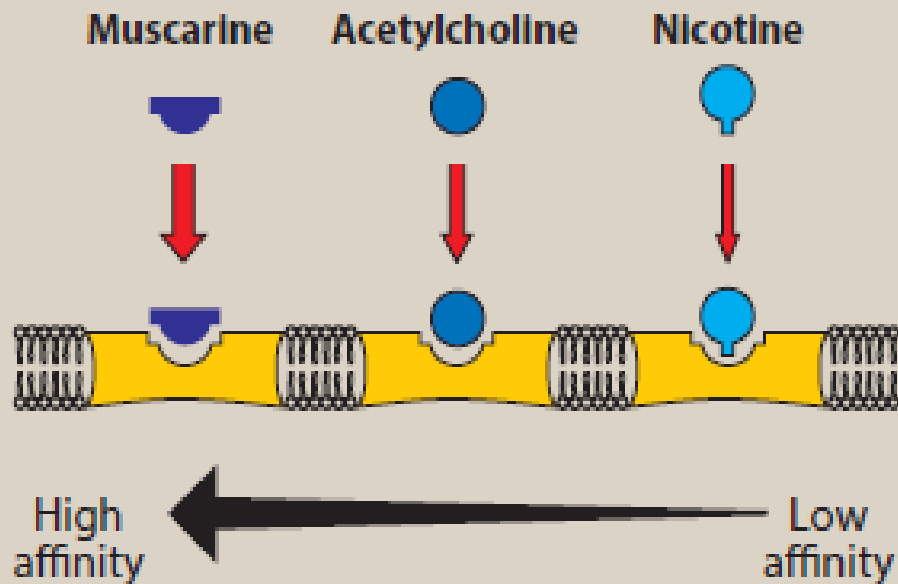
❑ Parasympathetic (cholinergic)

- Exit from cranial and sacral portions
- Neurotransmitter- acetylcholine
- Preganglionic neuron is longer than postganglionic neuron
- Cholinergic receptors- **N_N , N_M , $M1$, $M2$, $M3$, $M4$, & $M5$**

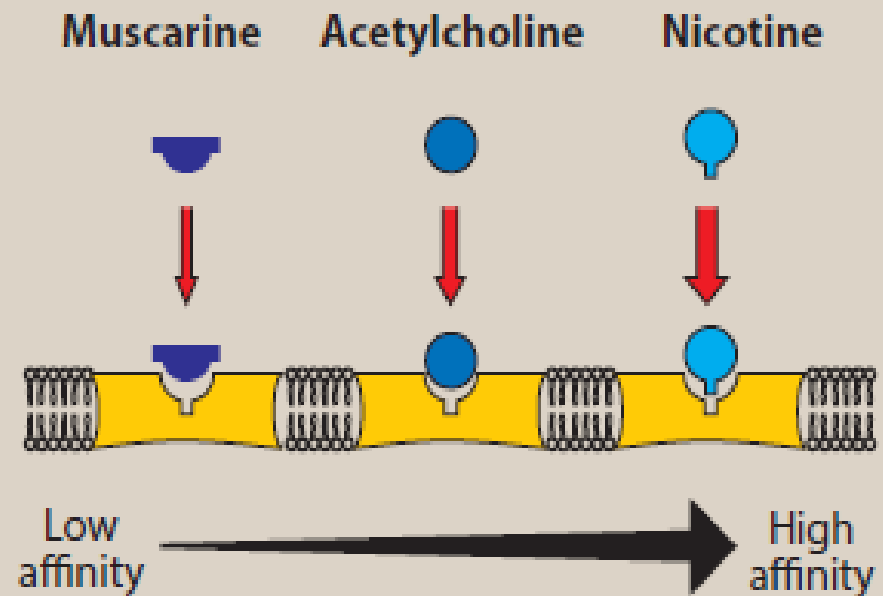


Types of cholinergic receptors:

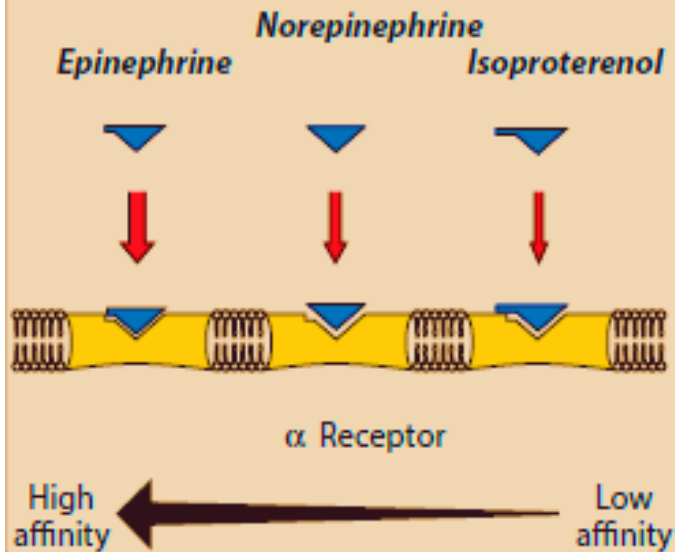
A Muscarinic receptors



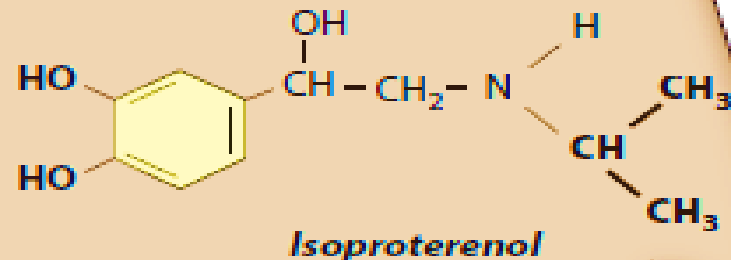
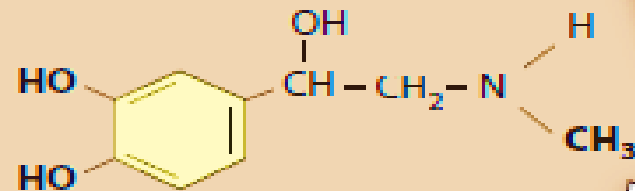
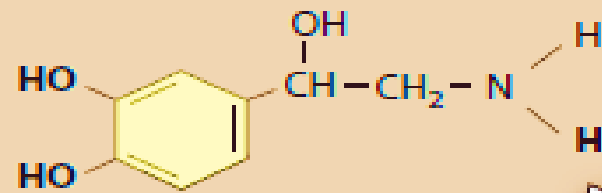
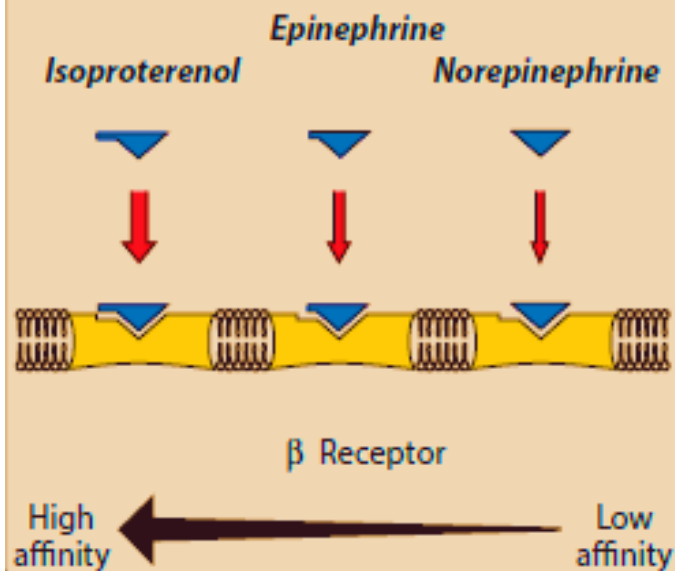
B Nicotinic receptors



A α -Adrenoceptors



B β -Adrenoceptors



Affinity for β receptors increases as group on the amine nitrogen gets larger.

Effects of ANS on innervated organs

Effector Organ	Sympathetic (NE)	Parasympathetic (Ach)
Heart	β_1 (Increased Heart Rate, force of contraction, Conduction Velocity)	M2 (Decreased HR, Force of contraction, Conduction velocity)
Lungs	β_2 (Bronchodilation)	M3 (Bronchoconstriction)
Arterioles of skin, mucosa & abdominal viscera, Veins	α_1 (Constriction)	-
Blood vessels of skeletal muscles and heart	β_2 (Dilation)	M (Dilation of cardiac BV)
GIT	Decreased GI Motility and tone with contraction of sphincters, α_1 ; Decreased secretions, α_2	M3 (Increased motility tone and secretion with relaxation of sphincters)
Male sex organ	α_1 (Ejaculation)	M3 (Erection)
Eye (Radial muscle of iris)	α_1 (Contraction- Mydriasis)	-
Eye (Sphincter muscle of iris)	-	M3 (Contraction- miosis)
Eye (Ciliary muscle)	β_2 (Relaxation - far vision)	M3 (Contraction - near vision)
Salivary gland	α_1 , Increased salivation (viscous)	M3 , Increased salivation (watery)

Effects of ANS on innervated organs

Effector Organ	Sympathetic (NE)	Parasympathetic (Ach)
Kidney	β_1 (increased renin release) α_1 (decreased renin release)	-
Urinary bladder	β_2 (relaxation of detrusor) α_1 (Contraction of trigone & sphincter) $\rightarrow$ retention of urine	M3 (contraction of detrusor) M3 (relaxation of trigone & sphincter) $\rightarrow$ Voiding
Pancreas	β_2 (increased insulin secretion) α_2 (decreased insulin secretion)	-
Uterus	β_2 (relaxation) α_1 (contraction, pregnant)	M3 (contraction)
Sympathetic nerve terminal	α_2 (Decreased NE release)	M2, M4 (decreased NE release)
Parasympathetic nerve terminal	α_2 (Decreased Ach release)	M2, M4 (decreased Ach release)

Classifications of drugs acting on ANS

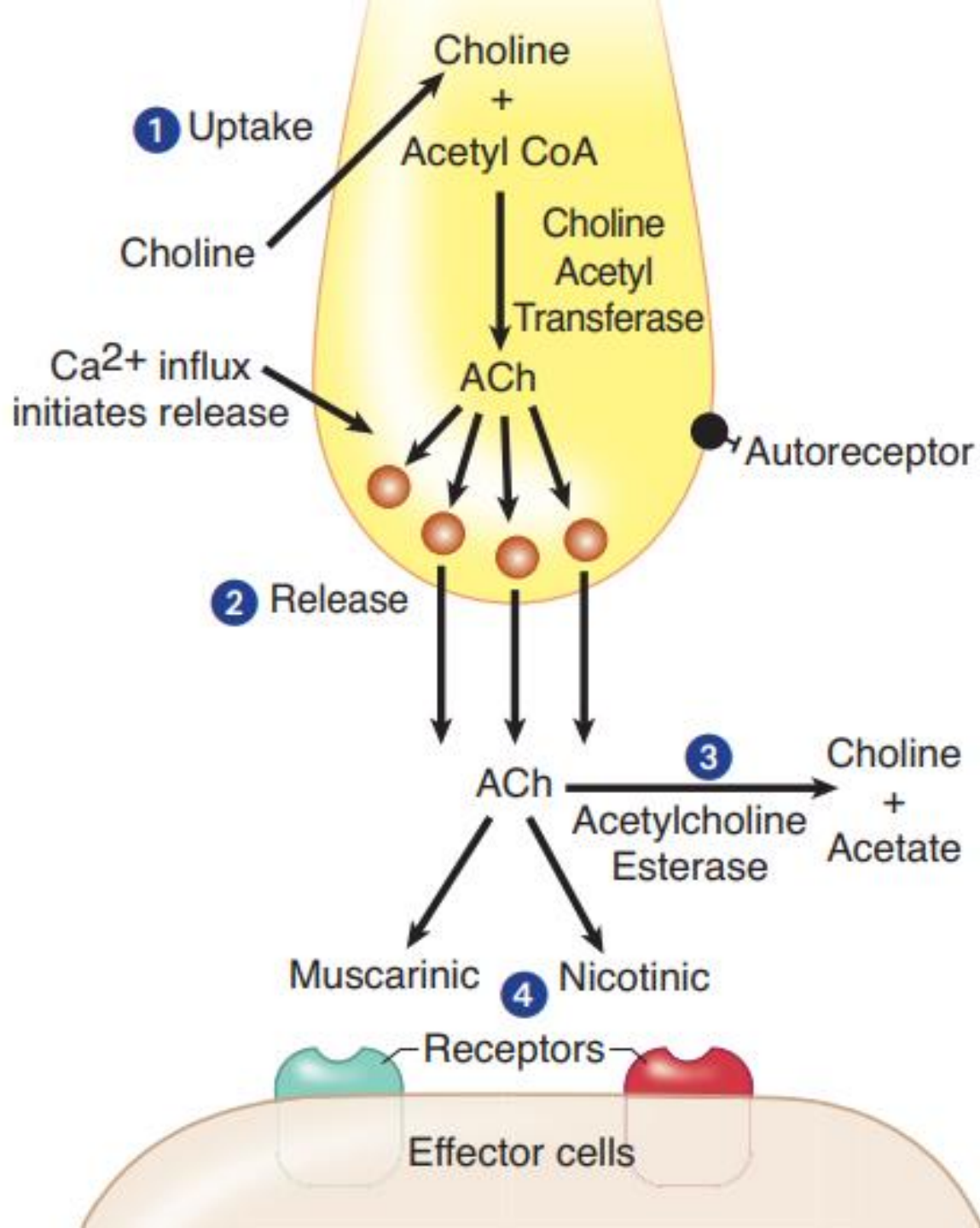
1. Cholinergic agonists (parasympathomimetics)
2. Cholinergic blockers (parasympatholytics)
3. Adrenergic agonists (sympathomimetics)
4. Adrenergic blockers (sympatholytics)

Direct-acting cholinergic agonists

- Classified into two groups:
 - **Choline esters**, which include Ach (endogenous choline ester) and synthetic esters of choline, such as *carbachol* and *bethanechol*
 - **Naturally occurring alkaloids**, such as *nicotine* and *pilocarpine*
- All of the direct-acting cholinergic drugs have
 - a longer duration of action than ACh.
- The more therapeutically useful drugs;
 - *pilocarpine* and *bethanechol*

Acetylcholine

- Released from Cholinergic (Parasympathetic) nerve terminal
- is a quaternary ammonium compound that cannot penetrate membranes.
- it lacks therapeutic importance because of
 - Its multiplicity of actions (activate both M & N receptors) and
 - its rapid inactivation by the Acetylcholine Esterase (AChE)



Properties of choline esters:

Choline Ester	Susceptibility to Cholinesterase	Muscarinic Action	Nicotinic Action
Acetylcholine chloride	++++	+++	+++
Methacholine chloride	+	++++	None
Carbachol chloride	Negligible	++	+++
Bethanechol chloride	Negligible	++	None

Bethanechol

- not hydrolyzed by AChE although it is inactivated through hydrolysis by other esterases.
- It **lacks nicotinic actions** but has strong muscarinic activity.
- Its major actions are on the smooth musculature of
 - bladder and GI tract.
- Therapeutic applications:
 - used to stimulate the atonic bladder, particularly in postpartum or postoperative, non-obstructive urinary retention
 - postoperative or postpartum intestinal atony

Cholinomimetic alkaloids

- Includes
 - **Pilocarpine**use externally
 - muscarine
 - arecoline
 - nicotine
- not clinically used due to toxicity
- 

Pilocarpine

- An alkaloid obtained from *Pilocarpus microphyllus* and *Pilocarpus japorandi*
- *a tertiary amine and* is stable to hydrolysis by AChE.
- Therapeutic use:
 - the **drug of choice for emergency lowering of intraocular pressure** of both open-angle and angle-closure glaucoma.
 - miotic action of *pilocarpine is also useful in reversing mydriasis due to muscarinic blockers*

How Pilocarpine treats glaucoma?

- Decrease in intraocular pressure due to two reasons
 - i. Ciliary muscle contraction puts tension on trabecular meshwork AND
 - ii. Contraction of the iris sphincter pulls the peripheral iris away from the trabecular meshwork



- Opening pores of trabecular meshwork and facilitating outflow of aqueous humor into the canal of Schlemm

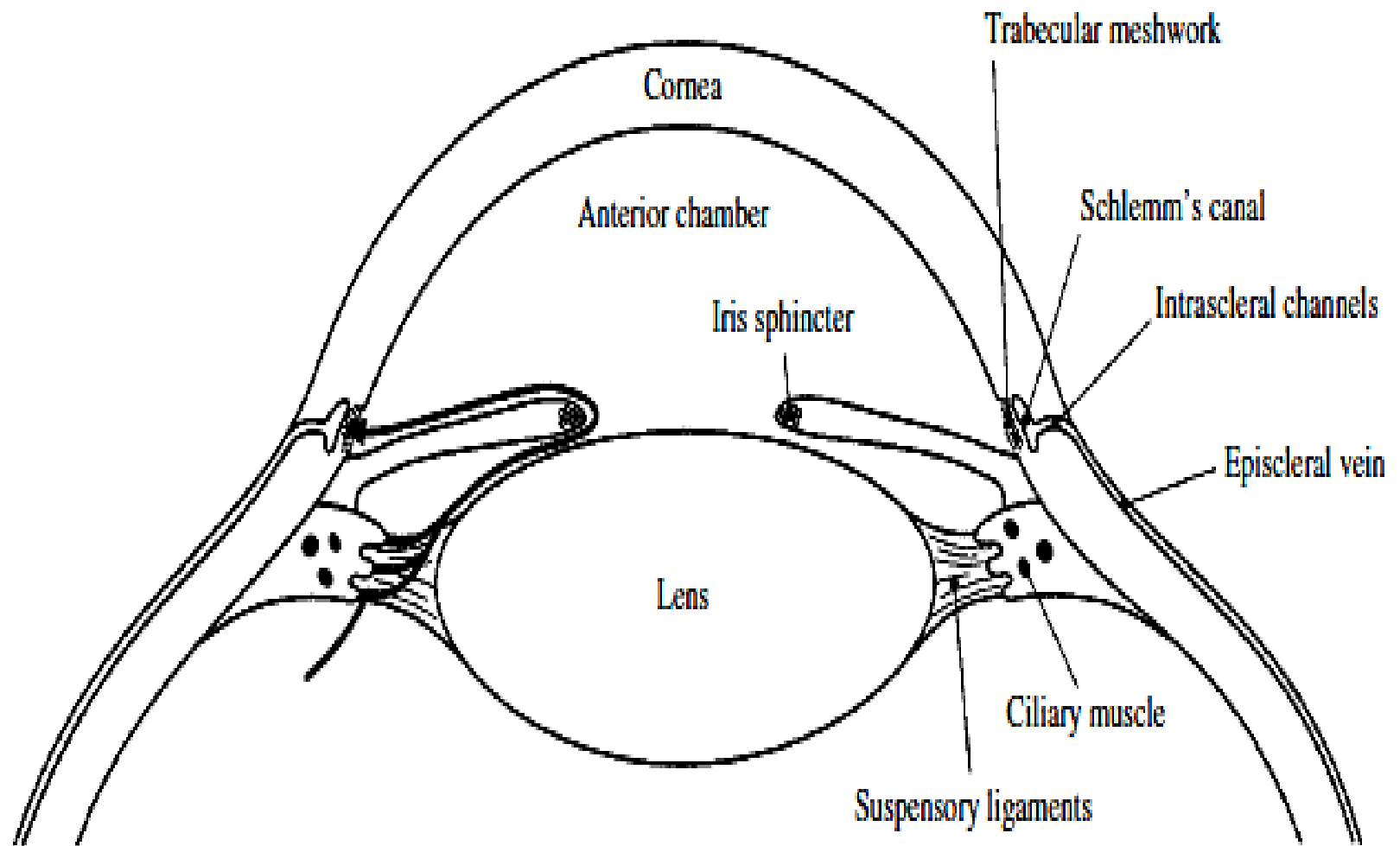


FIGURE 12.3

Diagram of eye depicting major pathway for outflow of aqueous humor (arrow) and ocular smooth muscles, which contract in response to parasympathomimetics or cholinesterase inhibitors (i.e., iris sphincter and ciliary muscle).

Glaucoma

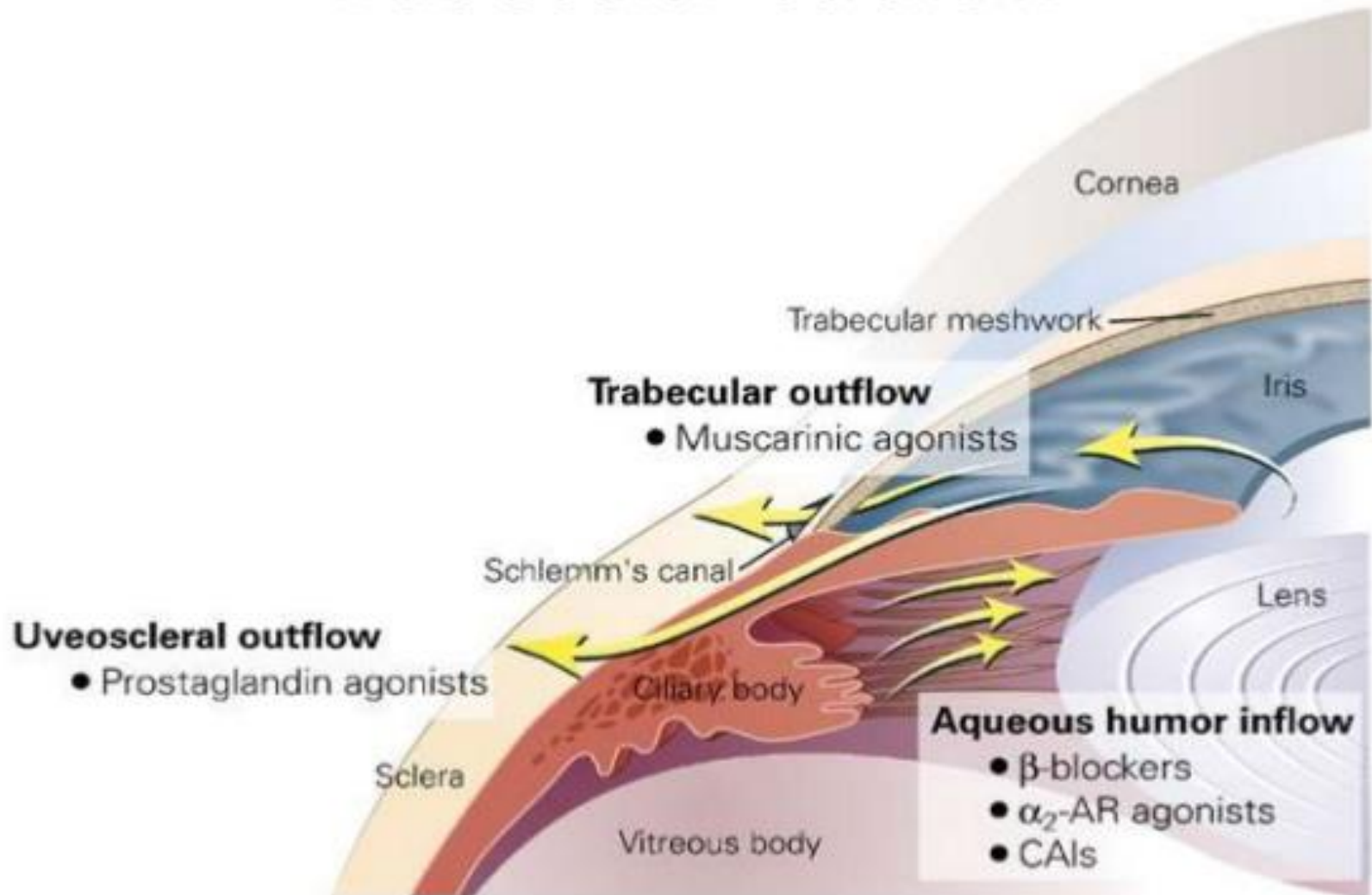
- Accumulation of fluid in the eye & increased intraocular pressure
→ optic nerve damage
- Could be

Open angle glaucoma

- ✓ has a gradual, insidious onset
- ✓ Porosity of the trabecular meshwork is not adequate to permit movement of fluid into the canal of Schlemm
- ✓ control of intraocular pressure usually is dependent on continuous drug therapy.

- Inflow of aqueous humor is decreased by
 - α 2-adrenergic agonists and β -adrenergic blockers and carbonic anhydrase inhibitors (CAIs)
- Outflow is increased by
 - **Muscarinic agonists**, which contract the ciliary muscle and open
 - the trabecular meshwork
 - **prostaglandin analogues** and α 2-adrenergic agonists,
 - enhance uveoscleral outflow
- ❖ **Prostaglandin analogues, β adrenergic receptor antagonists, and carbonic anhydrase inhibitors are**
 - first line drugs for open angle glaucoma

Trabecular outflow



Angle closure glaucoma

- ✓ Abnormal positioning of the peripheral iris blocking the access of fluid to the trabecular meshwork
- ✓ a medical emergency in which drugs are essential in controlling the acute attack, but Iridectomy (laser or surgical) is the definitive treatment
- ✓ Drug therapy of an acute attack typically involves administration of
 - **Pilocarpine, hyperosmotic agents, and a secretory inhibitor** (a β -blocker, α_2 -agonist, prostaglandin analog, or a topical or systemic CAI).

Indirect-acting cholinergic agonists

Cholinesterase inhibitors (Anticholinesterases)

- AChE is an enzyme that specifically cleaves ACh to acetate and choline and,
 - thus, terminates its actions.
- It is located both pre and postsynaptically in the nerve terminal where it is membrane bound.
- Inhibitors of AChE indirectly provide a cholinergic action by preventing the degradation of Ach

Cholinesterases inhibitors cont'd...

Reversible

- **Carbamates**
 - Physostigmine
 - Neostigmine
 - Pyridostigmine
 - Demecarium
 - Rivastigmine
- **Phenol**
 - Edrophonium

Irreversible

- **Organophosphates**
 - Diisopropyl fluorophosphates
 - Echothiophate
 - Parathion
 - Malathion
 - Diazinon (Tik-20)
 - **Tabun**
 - **Sarin**
 - **Soman**
- } Nerve gases for chemical warfare

Cholinesterase inhibitors (Reversible)

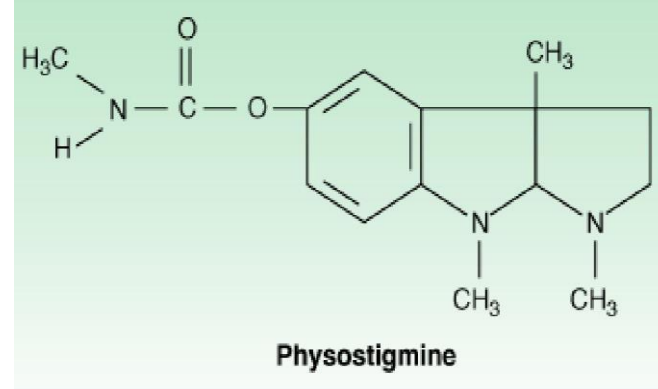
1) Alcohols

❑ Edrophonium

- Rapidly absorbed after parenteral admin/n, and has a short duration of action of
 - 10 to 20 minutes due to rapid renal elimination.
- *Edrophonium* is a quaternary amine
 - actions are limited to the periphery.
- It is used in the diagnosis of myasthenia gravis (differentiate b/n MG and cholinergic crisis)

2) Carbamates

☐ Physostigmine



- found naturally in plants and is a tertiary amine.
- Its duration of action is about 30 minutes to 2 hours.
 - it is considered an intermediate-acting agent.
- can enter and stimulate the cholinergic sites in the CNS.
- **Therapeutic uses:**
 - intestinal and bladder atony
 - overdoses of anticholinergic drugs,
 - ✓ such as atropine.

- **Adverse effects:**

- ON CNS, convulsions when high doses are used.
- Bradycardia and a fall in cardiac output may also occur.
- Inhibition of AChE at the skeletal NMJ may cause paralysis of skeletal muscle. However, rarely seen with therapeutic doses.

☐ **Neostigmine**

- Unlike physostigmine, neostigmine has a quaternary nitrogen.
 - is absorbed poorly from the GI tract, and
 - does not enter the CNS.
- Its effect on skeletal muscle is greater than that of physostigmine, and it can stimulate contractility before it paralyzes.

- **Therapeutic uses:**

- used to stimulate the bladder and GI tract
- an antidote for competitive neuromuscular junction blockers
- also used to manage symptoms of myasthenia gravis.

- ❑ **Pyridostigmine and ambenonium**

- are used in the chronic management of
 - myasthenia gravis.
- Their durations of action are intermediate (3 to 6 hours and 4 to 8 hours, respectively)
 - but longer than that of *neostigmine*.

❑ Tacrine, donepezil, rivastigmine, and galantamine

- *Tacrine was the first to become available, but it has been replaced by others*
 - because of its hepatotoxicity.
- Used to delay the progression of Alzheimer's disease,
 - none can stop its progression.
- Alzheimer's disease
 - deficiency of cholinergic neurons in the CNS.
- ***GI distress*** is their primary adverse effect

Cholinesterase inhibitors (Irreversible)

- A number of synthetic organophosphate compounds have the capacity to bind covalently to AChE.
- The result is a long-lasting increase in ACh at all sites where it is released.
- Many of these drugs are extremely toxic and
 - were developed for military purposes
- parathion and malathion, are used as insecticides.

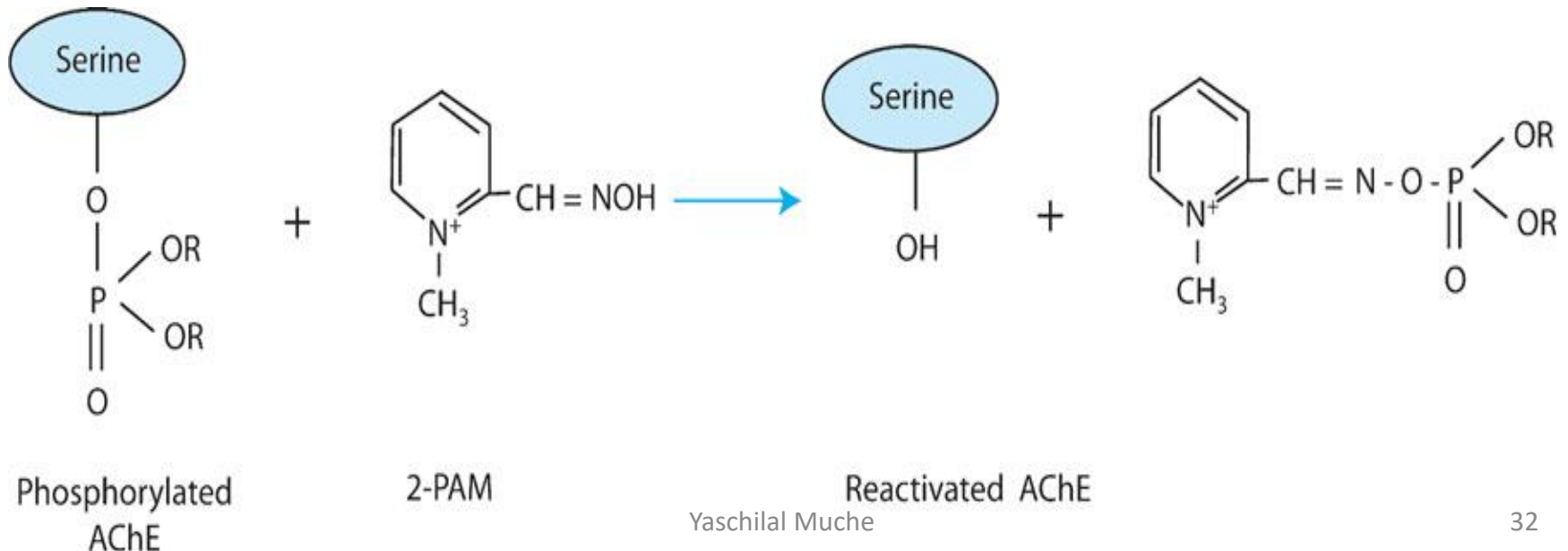
Organophosphates; irreversible inhibitors of AChE

- Tetraethylpyrophosphate
- Echothiophate
- Diisopropylfluorophosphate (DFP)
- Sarin
- Soman
- Tabun
- Malathion
- Parathion
- Diazinon
- Chlorpyrifos
- Many others

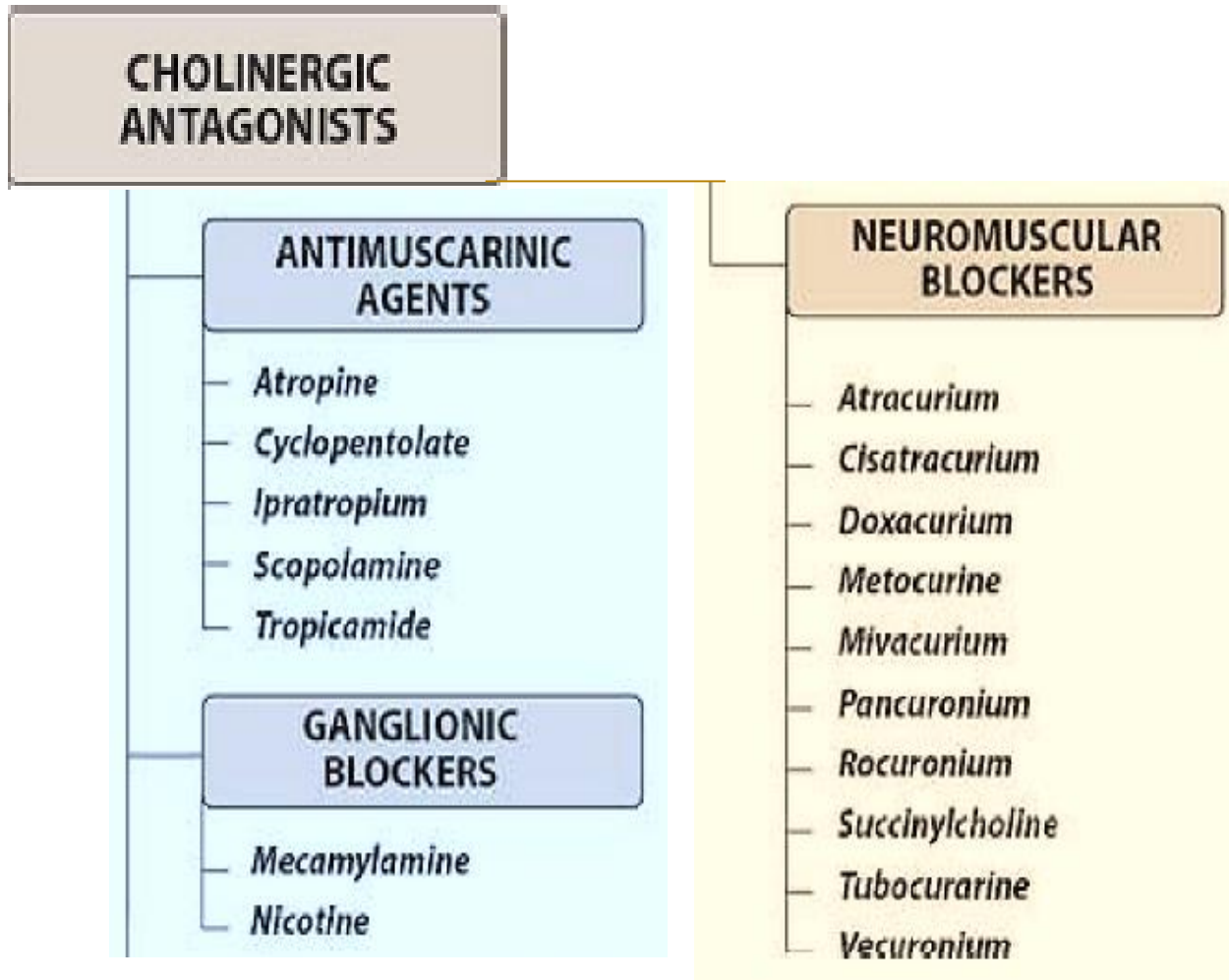
Cont....

Treatment of Organophosphate Poisoning;

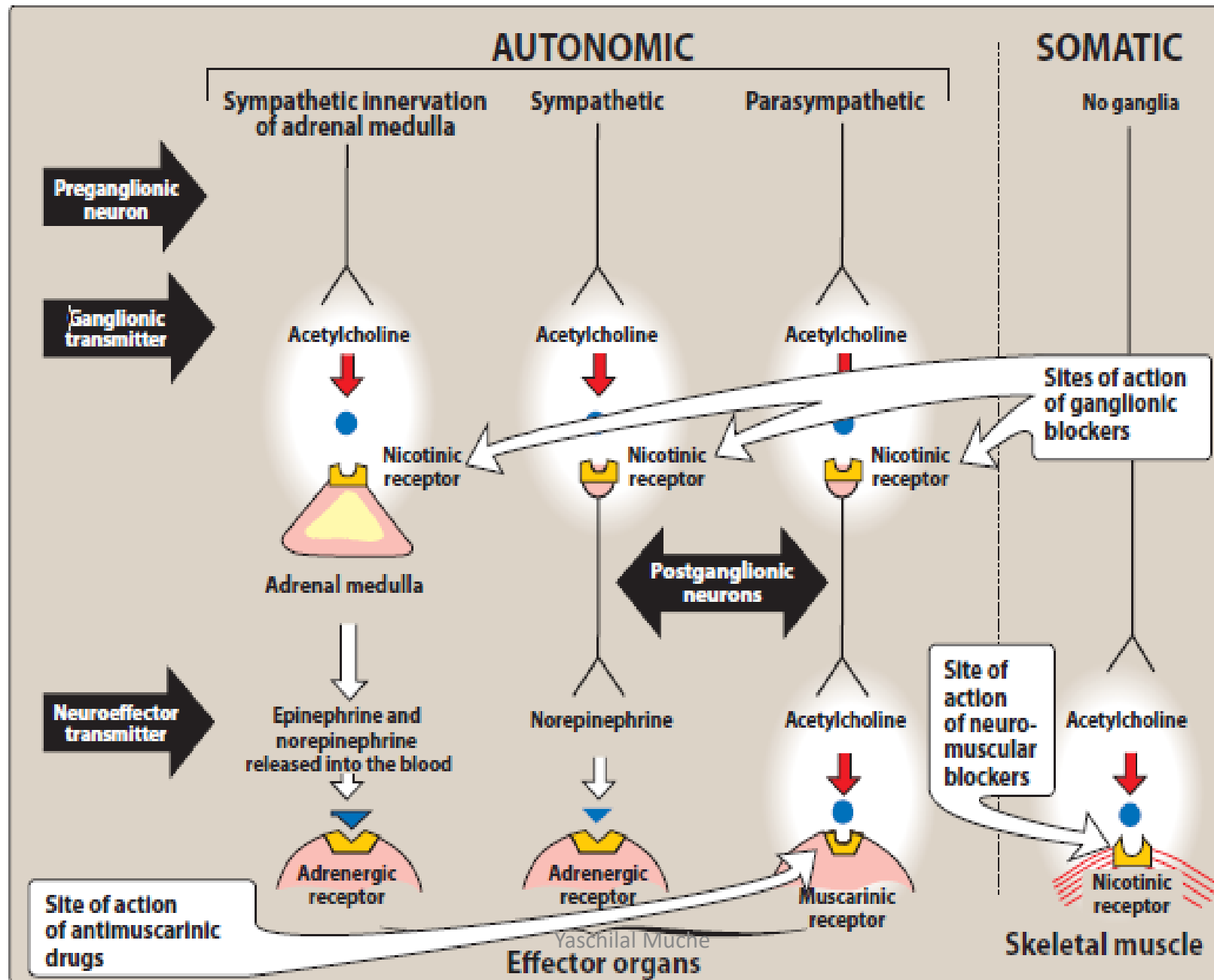
- Decontamination and/or minimizing absorption
- **Atropine** represents the cornerstone of the t/t for OP poisoning
- Oximes, such as pralidoxime (2-PAM) before aging



Cholinergic antagonists



Site of action of cholinergic antagonists;



Muscarinic blockers (Antimuscarinic agents)

- ☞ Block M but do not block the N receptors
- ☞ have no action at skeletal NMJ or autonomic ganglia

■ Classification

- ➡ Natural alkaloids: atropine, hyoscine (scopolamine)
- ➡ Semi-synthetic derivatives: homatropine, atropine methonitrate, hyoscine butyl bromide, ipratropium bromide, tiotropium bromide
- ➡ Synthetic compounds: Cyclopentolate, tropicamide, pirenzepine, telenzepine, biperiden, benztropine

1) Atropine

- *a tertiary amine belladonna alkaloid*
- *acts both centrally and peripherally.*

Actions:

- **Eye:** *blocks muscarinic activity in the eye, resulting in mydriasis* (dilation of the pupil), unresponsiveness to light, and cycloplegia (inability to focus for near vision).
- **GIT:** *Atropine and scopolamine* are probably the most potent antispasmodic drugs available.



- **CV:** low doses, the predominant effect is a slight decrease in heart rate.
 - results from blockade of the M1 receptors on the inhibitory presynaptic neurons, thus permitting increased ACh release.
- Higher doses of *atropine* cause a progressive increase in heart rate by blocking the M2 receptors on the sinoatrial node.
- **Secretions:** *Atropine blocks muscarinic receptors in the salivary glands, producing dryness of the mouth (xerostomia).*

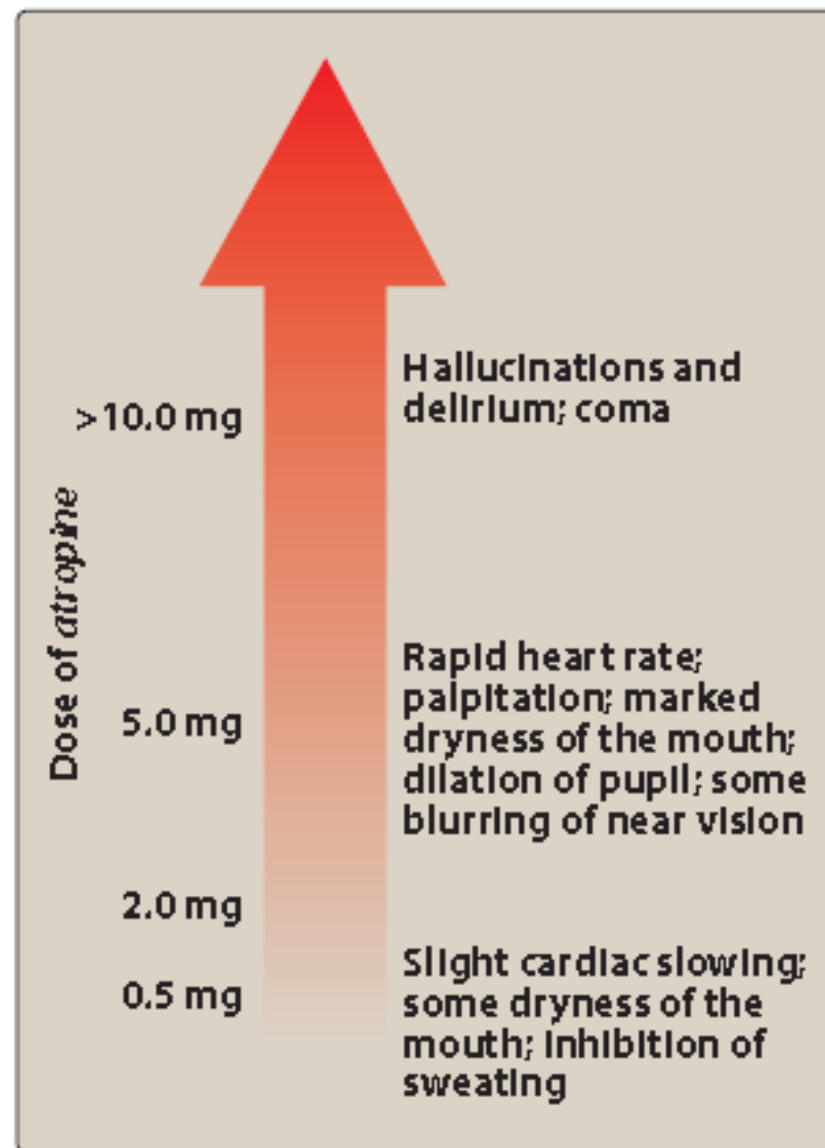


Figure 5.4

Dose-dependent effects of *atropine*.

Therapeutic uses of atropine:

- Ophthalmic: exerts both mydriatic and cycloplegic effects, and
 - it permits the measurement of refractive errors without interference by the accommodative capacity of the eye.
- Antispasmodic
- Cardiovascular (to treat bradycardia)
- Antisecretory; to block secretions in the upper and lower respiratory tracts prior to surgery
- Antidote for cholinergic agonists

Adverse effects:

- dry mouth, blurred vision, tachycardia, urinary retention, and constipation.
- Effects on the CNS include restlessness, confusion, hallucinations, and delirium, which may progress to
 - depression, collapse of the circulatory and respiratory systems, and death.

2) Scopolamine

- another tertiary amine plant alkaloid
- produces peripheral effects similar to those of atropine.
- However, scopolamine has greater action on the CNS
 - unlike atropine, CNS effects are observed at therapeutic doses and
 - a longer duration of action as compared to atropine.



Actions:

- One of the most effective anti-motion sickness drugs available.
- has the unusual effect of **blocking short-term memory**.
- In contrast to atropine, scopolamine produces sedation,
 - but at higher doses, it can produce excitement.
- Scopolamine may produce euphoria and is susceptible to abuse.

Therapeutic uses:

- prevention of motion sickness and postoperative nausea and vomiting.

3) **Ipratropium and tiotropium**

- are quaternary derivatives of *atropine*.
- *are* approved as bronchodilators for maintenance treatment of bronchospasm associated with COPD.
- *Ipratropium* is also used in the acute management of bronchospasm in asthma

4) **Benztropine (trihexyphenidyl)**

- *Adjunct therapy* for Parkinson's disease and other types of parkinsonian syndromes, including
 - antipsychotic-induced extrapyramidal symptoms.

5) Tropicamide and cyclopentolate

- used topically to induce mydriasis and cycloplegia.
- Their duration of action is shorter than that of atropine.
 - Tropicamide produces mydriasis for 6 hours, and cyclopentolate for 24 hours.

6) Darifenacin, fesoterodine, oxybutynin, solifenacin, tolterodine, and trospium chloride

- synthetic *atropine*-like drugs
- used to treat overactive bladder.

Neuromuscular blocking drugs

➡ Block cholinergic transmission b/n

➤ motor nerve endings and the nicotinic receptors on the neuromuscular endplate of skeletal muscle.

➡ Structural analogue of Ach

➡ Useful clinically during surgery

➤ to produce complete muscle relaxation

Neuromuscular blocking drugs ...

A. Competitive (Non depolarizing) blockers

- **D-tubocurarine**
- Doxacurium
- Gallamine
- Vecuronium
- Pancuronium
- Rocuronium
- Alcuronium
- Mivacurim
- Atracurium

B. Depolarizing blockers

- Suxamethonium
(**succinylcholine**)
- Decamethonium

1. Non-depolarizing (competitive) blockers



- ➡ The first drug was **curare**,
 - Has been used by native hunters of the Amazon to paralyze animals.
- ➡ **Tubocurarine** was purified and introduced into clinical practice
- ➡ but Tubocurarine was replaced by other agents with fewer adverse effects,
 - *cisatracurium, pancuronium, rocuronium, and vecuronium.*

Non-depolarizing (competitive) blockers ...

➡ Mechanism of action

❑ At low doses:

- Competitively block ACh at the nicotinic receptors
- Prevent depolarization of the muscle
 - causing flaccid paralysis that lasts 30-60 minutes (longer in large doses)
- Their action can be overcome by increasing the concentration of ACh in the synapse
 - AChE inhibitors such as neostigmine or edrophonium

Non-depolarizing (competitive) blockers ...

➔ Mechanism of action..

❑ At high doses:

- can block the ion channels of the motor endplate
- This leads to further weakening of neuromuscular transmission, thereby
 - ✓reducing the ability of cholinesterase inhibitors to reverse the actions of the non-depolarizing blockers

☞ Therapeutic uses:

- Adjuvant drug in anesthesia during surgery to relax skeletal muscle

Non-depolarizing (competitive) blockers ...

➡ PK

- All NMJBs administered IV or occasionally IM
- Oral absorption is minimal (two or more quaternary amines)
- Most of them are excreted primarily unchanged in urine

➡ AEs:

- Safe with minimal SE

Drug interactions:

- **Cholinesterase inhibitors:** Neostigmine, physostigmine, pyridostigmine, and edrophonium can
 - overcome the action of non-depolarizing neuromuscular blockers.
- **Halogenated hydrocarbon anesthetics:** sensitize the NMJ to the effects of NMJBs
- **Aminoglycoside antibiotics:** enhance the blockage
- **Calcium channel blockers:** enhance the blockage

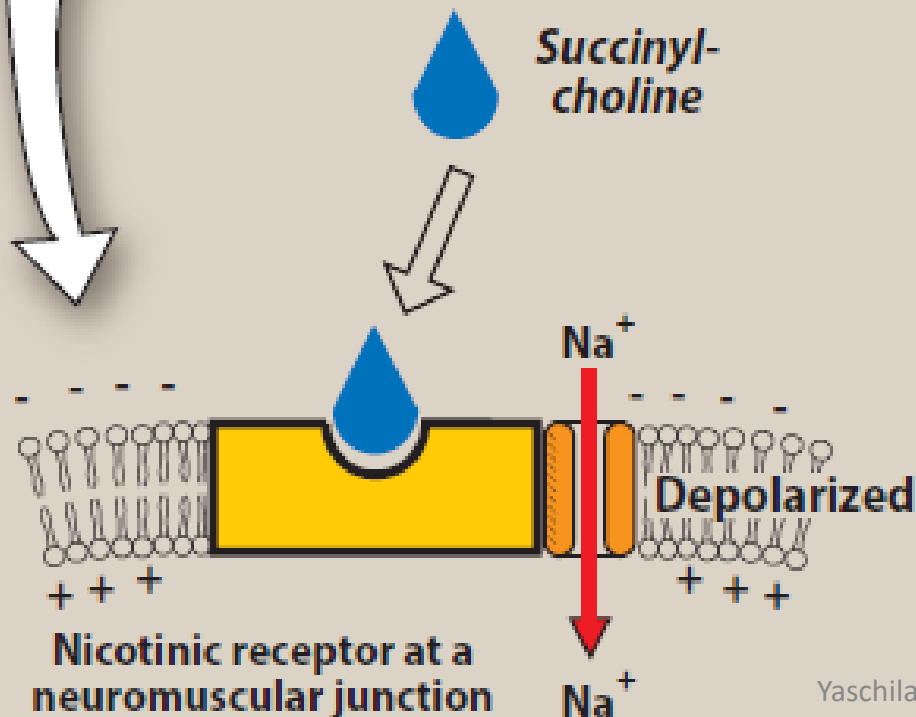
2) Depolarizing NMJBs

- work by depolarizing the plasma membrane of the muscle fiber, similar to the action of ACh.
 - However, these agents are more resistant to degradation by AChE
- **Succinylcholine** is the only depolarizing muscle relaxant in use today.

Mechanism of action of depolarizing NMJBs

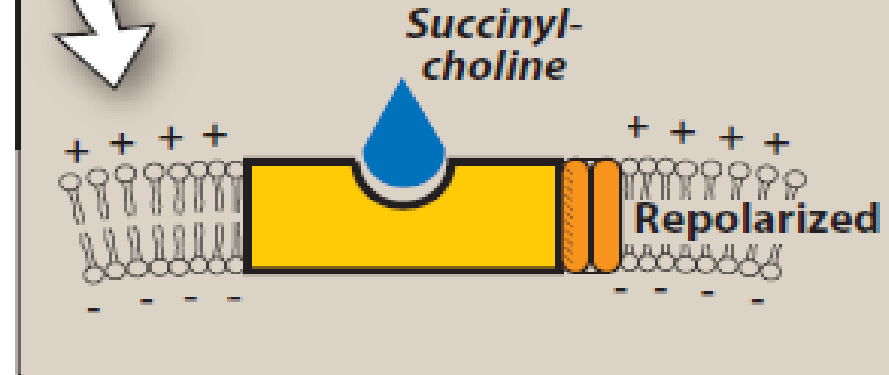
PHASE I

Membrane depolarizes, resulting in an initial discharge that produces transient fasciculations followed by flaccid paralysis.



PHASE II

Membrane repolarizes, but receptor is desensitized to the effect of acetylcholine.



- The duration of action of *succinylcholine* is dependent on diffusion from the motor endplate and hydrolysis by
 - **plasma pseudocholinesterase.**
 - Genetic variants in which plasma **pseudocholinesterase** levels are low or absent lead to prolonged neuromuscular paralysis.

Depolarizing agents ...

Therapeutic uses:

- ➡ Because of its rapid onset of action, succinylcholine is useful **when rapid endotracheal intubation** is required during the induction of anesthesia (a rapid action is essential if aspiration of gastric contents is to be avoided during intubation).
- ➡ It is also used during electroconvulsive shock treatment

AEs:

- **Malignant hyperthermia:** halogenated hydrocarbon anesthetics + succinylcholine (t/t - **dantrolene**)
- **Apnea:** in patients with plasma pseudocholinesterase deficiency, due to paralysis of the diaphragm.
- **Hyperkalemia:** increases K⁺ release from intracellular stores

Adrenergic (Sympathomimetic) drugs

Classification

✓ Based on their chemical structures

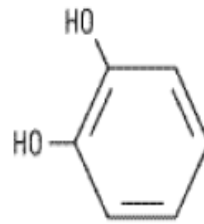
1. Catecholamines:

- **Natural:** Adrenaline, Noradrenaline, Dopamine
- **Synthetic:** Isoprenaline, Dobutamine

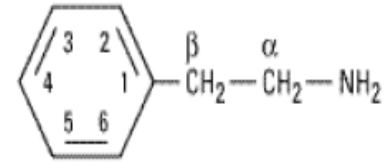
2. Non-Catecholamines:

- Ephedrine, Amphetamines, Phenylephrine

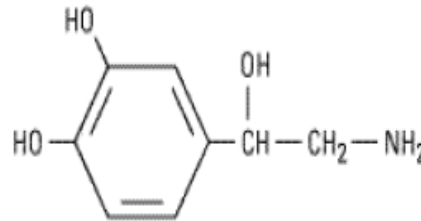
- **Catecholamines:** Cmpds containing a *catechol nucleus* (Benzene ring with 2 adjacent OH groups) and an *amine* containing side chain
- **Non-catecholamines** lack hydroxyl group



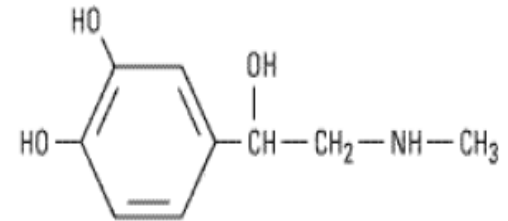
Catechol



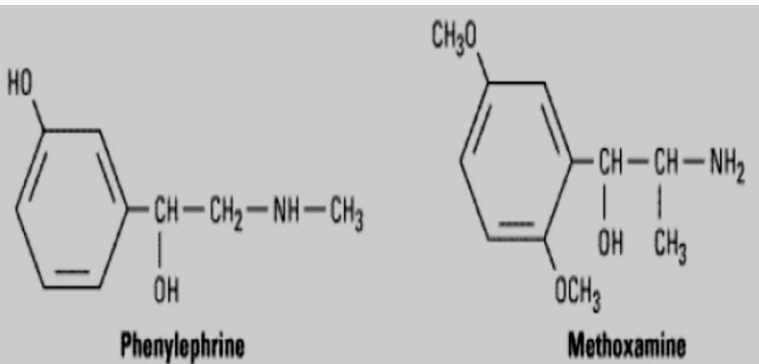
Phenylethylamine



Norepinephrine

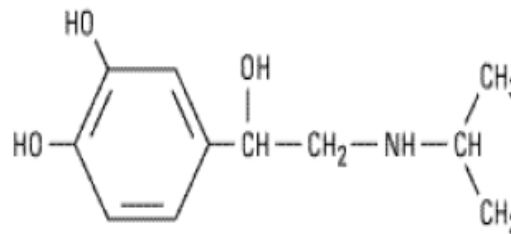


Epinephrine

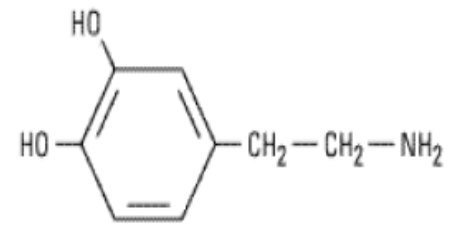


Phenylephrine

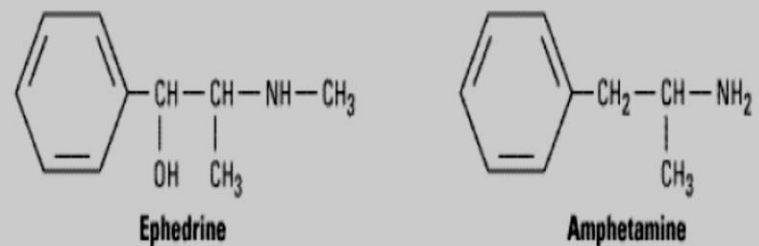
Methoxamine



Isoproterenol



Dopamine



Ephedrine

Amphetamine

Catecholamines;

- show the highest potency in directly activating α or β receptors
- Rapid inactivation: metabolized by COMT postsynaptically and by MAO intraneuronally, as well as by COMT and MAO in the gut wall, and by MAO in the liver.
- Catecholamines are polar and, therefore, do not readily penetrate into the CNS.

Noncatecholamines;

- have longer halflives
 - they are not inactivated by COMT
 - are poor substrates for MAO
- Increased lipid solubility (due to lack of polar hydroxyl groups) permits greater access to the CNS.

Mechanism of action of adrenergic agonists;

1. Direct-acting agonists: These drugs act directly on α or β receptors.

➤ *epinephrine, norepinephrine, isoproterenol, and phenylephrine.*

2. Indirect-acting agonists: These agents may block the reuptake of NE, or cause the release of NE from the cytoplasmic pools or vesicles of the adrenergic neuron, or inhibit the metabolism of NE.

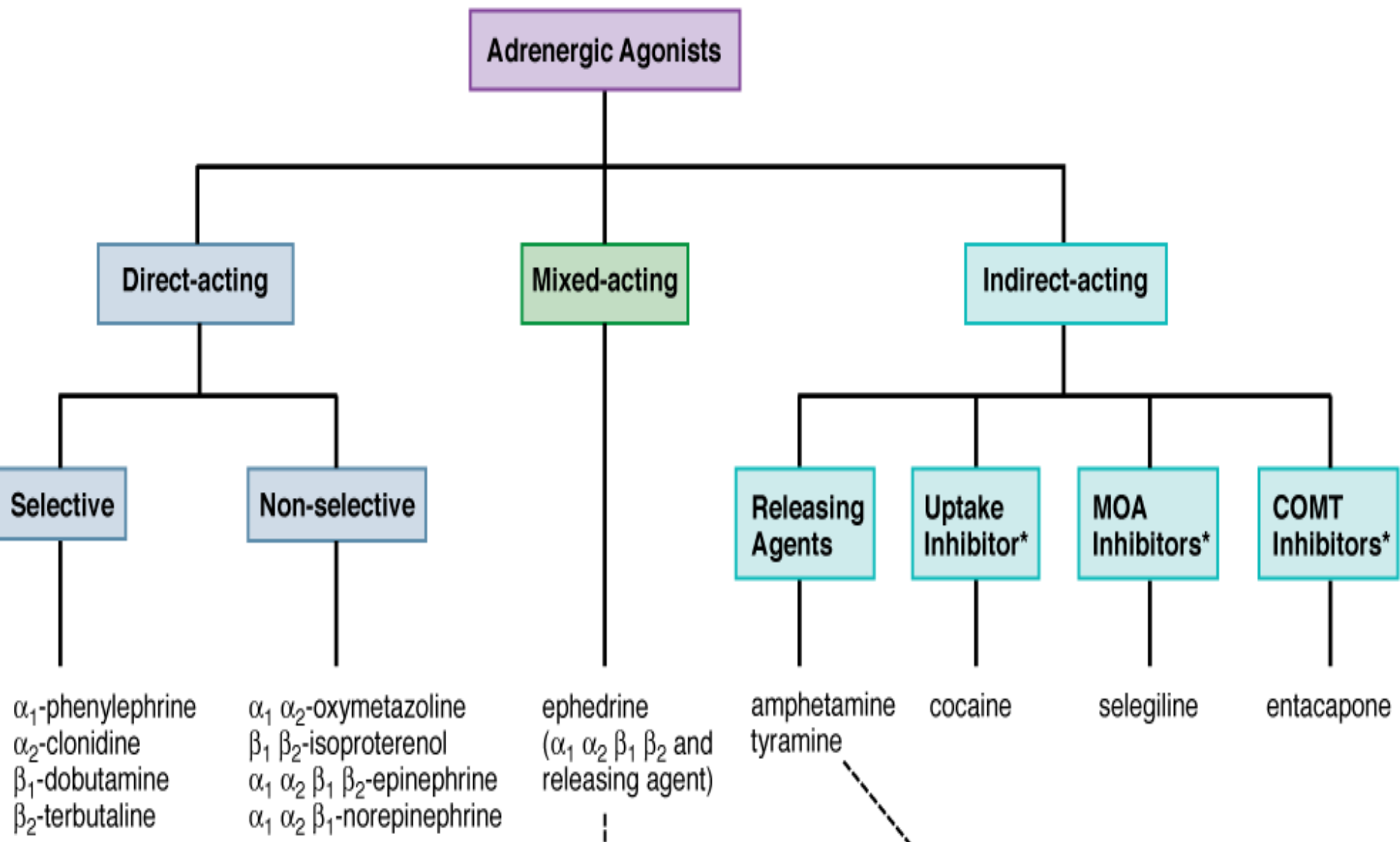
➤ Examples; NE reuptake inhibitors (*cocaine, TCADs*) and agents that cause norepinephrine release (*amphetamines*)

3. Mixed-action agonists: Ephedrine and pseudoephedrine

➤ stimulate adrenoceptors directly and

➤ Stimulate norepinephrine release from the adrenergic neuron





Adrenergic (sympathomimetic) drugs

<u>Drugs</u>	<u>Receptor activated</u>
isoproterenol	B1 and B2
adrenaline	α 1, α 2, B1 and B2
Noradrenaline	α 1, α 2 and B1
dopamine	D1, D2, B1
dobutamine	B1
ephedrine	α 1, α 2, B1 and B2
Phenylephrine	α 1
Terbutaline, salbutamol	B2

Adrenaline as prototype

- Potent stimulant of α and β receptors

Actions on target organs

- Heart
 - Direct myocardial stimulation- +ve inotropic
 - Increased heart rate -+ve chronotropic
 - Increased conduction velocity- +ve dromotropic

Adrenaline

- Blood vessels
 - Vasoconstriction in skin, mucus membranes and viscera – α -1 effect
 - Vasodilation in skeletal muscles- (β -2 effect)
- ❖ At low doses, β effects (vasodilation) on the vascular system predominate, whereas at high doses, α effects (vasoconstriction) are the strongest.

Adrenaline...

Low dose

- β_1 : $\uparrow$ HR, $\uparrow$ SV, $\uparrow$ CO, $\uparrow$ pulse pressure
- β_2 : $\downarrow$ TPR, $\downarrow$ BP

Medium dose

- β_1 : $\uparrow$ HR, $\uparrow$ SV, $\uparrow$ CO, $\uparrow$ pulse pressure
- β_2 : $\downarrow$ TPR, $\downarrow$ BP
- α_1 : $\uparrow$ TPR, $\uparrow$ BP

High dose

- α_1 : $\uparrow$ TPR, $\uparrow$ BP
- Potential reflex bradycardia
- β_1 : $\uparrow$ HR, $\uparrow$ SV, $\uparrow$ CO, $\uparrow$ pulse pressure
- β_2 : $\downarrow$ TPR, $\downarrow$ BP

Adrenaline.....

Respiratory

- Powerful bronchodilator
 - β -2 mediated effect
 - It also inhibits the release of allergy mediators such as histamines from mast cells.

Smooth Muscles

- Effects on vascular smooth muscle are important
- GIT and Urinary tract smooth muscle are relaxed but are clinically unimportant
- In the uterus, there is inhibition of tone and contraction

Adrenaline.....

Metabolic effects

- **Hyperglycemia:** increased glycogenolysis in the liver (β_2 effect), increased release of glucagon (β_2 effect), and a decreased release of insulin (α_2 effect)
- **Lipolysis:** agonist activity on the β receptors of adipose tissue.
 - Increased levels of cAMP stimulate a hormone-sensitive lipase, which hydrolyzes triglycerides to free fatty acids and glycerol

Adrenaline.....

Pharmacokinetics

- Route of administration: IV, IM & SC
- Rapid onset but short duration
- Rapidly inactivated by MAO and COMT
 - Inhalation is locally effective
- Ineffective orally

Adrenaline.....

Clinical uses

- **Anaphylactic shock**- due to its effect on the CVS & on the bronchial muscles.
- Prolong action of local anesthetics
- In acute attack of bronchial asthma
- Cardiac arrest & complete heart block
- topical *hemostatic agent*- control of local hemorrhage

Adrenaline.....

Adverse effects

- Cerebral hemorrhage
- Cardiac arrhythmias
- precipitation of angina pectoris
- CNS disturbances: anxiety, fear, tension, headache, and tremor.
- Pulmonary edema

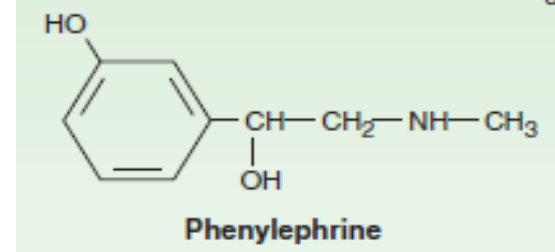
Note; *Epinephrine may have enhanced cardiovascular actions in patients* with hyperthyroidism, and the dose must be reduced in these individuals.

- Contraindication (CI)
 - In patients with severe hypertension, tachyarrhythmias, coronary artery disease.
- Drug interaction:
 - Inhalation anesthetics sensitize the heart to the effects of *epinephrine*, which may lead to tachycardia.
 - *Epinephrine causes hyperglycemia; thus*, In diabetic patients, dosages of *insulin/antidiabetics may have to* be increased.
 - Nonselective β -blockers prevent vasodilatory effects of *epinephrine on β_2 receptors*, leaving *α receptor stimulation unopposed* → increased TPR and increased BP.

Alpha adrenergic agonists

- Selective Alpha-1 Agonists
 - Phenylephrine, Methoxamine, Metaraminol
- Selective Alpha-2 Agonists
 - Clonidine, Apraclonidine, α -methyldopa, Guanfacine, Guanabenz
- Mixed alpha agonists
 - ✓ Oxymetazoline

Phenylephrine



- Synthetic, Selective and direct α_1 -agonist
- Administered parenterally & topically (eye, nose)
- Produces **mydriasis** and **nasal decongestion**
- Long duration of action
 - Resistant to **MAO** and **COMT**
- Peripheral vasoconstriction leads to **increased BP**
 - Reflex bradycardia

Phenylephrine...

- Does not cross BBB (has one hydroxyl group on the benzene ring), so no CNS effects
- Use- to treat hypotension in hospitalized or surgical patients (especially those with a rapid heart rate)
 - Sinusitis & Rhinitis as nasal decongestant
 - Mydriatic in the form of eye drops and lowers intraocular pressure
- **ADRs:** hypertension (systemic), Photosensitivity (topical)

Methyl dopa

MOA

- ① α methyl dopa is the substrate for Dopa decarboxylase and converted to α methyl dopamine & α -methyl norepinephrine
- ② α -methyl norepinephrine is powerful stimulator of α_2 & hence decrease release of NE
- ③ Decrease sympathetic out flow ($\downarrow$ HR & contractility, veins dilate ($\downarrow$ EDV)= $\downarrow$ SBP)
- ④ artery dilate ($\downarrow$ TPR= $\downarrow$ DBP)
- ⑤ JGA ($\downarrow$ renin, less Ag II)= less salt & water retention, veins & artery dilated

Methyl dopa....

➡ Mainly used for management of HTN in pregnancy

Side effects: sedation and drowsiness

- Dry mouth (xerostomia)
- Impotence (↓ sympathetic out flow)

Oxymetazoline

➔ Direct-acting α -agonist: stimulates both α_1 - and α_2 -adrenergic receptors.

Use:

- short-term decongestants (nasal spray)
- ophthalmic drops for the relief of redness of the eyes associated with swimming, colds, and contact lenses.

⊗ **Adverse effects** include: nervousness, headache, and trouble sleeping.

- Rebound congestion and dependence are observed with long-term use.

Beta - Adrenergic agonists

- B1 adrenergic agonists
 - Dopamine
 - Dobutamine- synthetic analogue of dopamine
- B2 Adrenergic agonists
 - Salbutamol
 - Terbutaline
 - Salmeterol

Dopamine

- Immediate metabolic precursor of Noradrenaline
- High concentration in basal ganglia, limbic system and hypothalamus of the CNS
- **ADME**- ineffective orally, IV use only
 - Short $t_{1/2}$ (3-5 minutes)

- **In small doses (2-5 μ g/kg/minute)**
 - It stimulates D1-receptors in renal, mesenteric and coronary vessels leading to vasodilation
- Clinically useful in the treatment of **shock** in which significant increase in sympathetic activity might compromise renal function.
 - Increases renal blood flow, GFR and causes natriuresis

- **Moderate dose (5-10 $\mu\text{g/kg/minute}$)**
 - stimulates β_1 -receptors in heart producing positive inotropic and chronotropic actions
 - Does not change TPR
- used in **Acute decompensated heart failure and shock**
 - **Dobutamin** - Preferred for acute decompensated heart failure
- **High dose (10-30 $\mu\text{g/kg/minute}$)**
 - stimulates vascular adrenergic α_1 -receptors
 - vasoconstriction and decreased renal blood flow

β_2 Adrenergic Agonists

- Short acting: Salbutamol, Metaproterenol, Terbutaline
 - Selective for β_2 receptor subtype
 - Used for acute inhalational treatment of bronchospasm.
 - Directly relax airway smooth muscle
 - Onset of action; 1 to 5 minutes
 - DOA; 2 to 6 hours
 - Duration of action is longer on oral administration
- Long acting: Salmeterol, formeterol

Mixed-action adrenergic agonists

Ephedrine

- ☞ A plant alkaloid, is now made synthetically
- ☞ Releases stored norepinephrine from nerve endings and directly stimulate both α and β receptors
- ☞ Ephedrine is not a catechol and poor substrate for COMT and MAO;
 - thus, the drug has a long duration of action but less potent
- ☞ Excellent absorption orally & penetrates in to the CNS
- ☞ Raises BP by **vasoconstriction** and **cardiac stimulation**
- ☞ Produces **bronchodilation**

- ☞ Used to treat **asthma**, as **nasal decongestant**, and to **raise BP**
 - The clinical use of ephedrine is declining due to availability of better, more potent agents with fewer side effects

Sympatholytics (Antiadrenergic) Drugs

↪ Adrenoceptor antagonists **classified based on their selectivity as:**

- ① Alpha – adrenoceptor blocking agents
- ② Beta – adrenoceptor blocking agents
- ③ Nonselective Alpha and beta adrenoceptor blocking agents

Alpha Receptor Antagonists

```
graph TD; A[Alpha Receptor Antagonists] --> B[Non-selective]; A --> C["α1-selective"]; A --> D["α2-selective"]
```

Non-selective

- phenoxybenzamine
- phentolamine

α₁-selective

- prazosin
- terazosin
- doxazosin
- alfuzosin
- tamsulosin
- indoramin
- urapidil
- bunazosin

α₂-selective

- yohimbine

Selective α_1 -antagonists

Includes: prazosin, terazosin, doxazosin, alfuzosin, tamsulosin

Prazosin:

- Cause both arterial and venous dilation, reducing peripheral vascular resistance and BP
 - Used to treat hypertension
 - Effect on receptors of prostate gland and bladder decreases resistance to urinary outflow
 - reducing urinary obstruction and relieving effects of BPH
- (Tamsulosin is preferred)**

- The α_1 receptor is further divided into
 - **α_1A , α_1B , α_1C , and α_1D subtypes.**
- *Tamsulosin is a selective α_1A antagonist that is used to treat benign prostatic hyperplasia.*
 - The drug has fewer cardiovascular side effects because it targets α_1A subtype receptors found primarily in the urinary tract and prostate gland and does not affect the α_1B subtype found in the blood vessels.
 - Except tamsulosin, others are non-selective among α_1 -receptor subtypes

Selective α_2 -antagonists

Mirtazapine: used as antidepressant

Yohimbine

- Is an alkaloid obtained from plants
- Readily enters to CNS
- Is competitive α_2 -selective antagonist
 - Increases sympathetic outflow
 - Increases blood pressure & heart rate
 - Produces opposite effects to clonidine

Therapeutic uses

- The treatment of **male erectile dysfunction (ED)**
- Not widely used due to availability of effective agents

Non-selective α -blockers

Phentolamine

- Quickly reverses the potent vasoconstrictive effects of extravasated vasopressors such as norepinephrine or epinephrine
- Restores blood flow and prevents tissue necrosis
- Used to control and prevent hypertension in patients
 - with pheochromocytoma

α -Blockers: Adverse Effects

Body System

Adverse Effects

Cardiovascular

Orthostatic hypotension, reflex tachycardia, edema

CNS

Dizziness, headache, drowsiness,
anxiety, depression, vertigo, weakness,
numbness, fatigue

Other

urinary incontinence, nasal bleeding,
tinnitus, dry mouth, pharyngitis,
rhinitis

β -Blockers

- Block stimulation of β receptors of the SNS
- Compete with norepinephrine and epinephrine
- **Selective** and **nonselective** β -blockers
 - Nonselective β -blockers block both β_1 and β_2 receptors

Examples;

- ❖ *Atenolol*
- ❖ *Metoprolol*
- ❖ *Bisoprolol*
- ❖ *Nebivolol*
- ❖ *Acebutolol*
- ❖ *Esmolol*

Selective β_1 blocker

- **Sotalol***
- **Timolol**
- **Nadolol**
- **propranolol**

Non-selective β blocker

- **Carvedilol**
- **labetalol**

Non-selective β & α_1 blockers

* K^+ -channel blockade and beta-blocking activity

Shortest half life: Esmolol (10 min)

β Receptors

B_1 receptors;

- Located primarily on the heart
 - B-blockers selective for these receptors are called cardioselective BBs
 - **MOA** - Reduces SNS stimulation of the heart
 - Decreases heart rate
 - Slows conduction rate through the AV node
 - Decreases myocardial contractility, thus reducing myocardial oxygen demand

β Receptors (cont'd)

B₂ receptors

- Located primarily on
 - **smooth muscles of bronchioles and**
 - **blood vessels of the skeletal muscle**

Nonselective blockers (**B₁** and **B₂**)

- Effects on heart: Same as cardioselective
- Bronchioles: Constriction, resulting in narrowing of airways and shortness of breath
- Blood vessels: Vasoconstriction

Indications of beta-blockers

- Angina (Ischemic heart disease): Decreases demand for myocardial oxygen
- Cardiac arrhythmia
 - Decrease automaticity and AV node conduction velocity
- Antihypertensive (decrease cardiac output & renin release)
- Some are used to treat chronic heart failure (**Carvedolol, metoprolol & Bisoprolol**)
- Treatment of migraine headaches (propranolol)
 - Blockade of cranial beta receptors reduce vasodilation

- Anxiety : to ↓ sympathetic manifestations (propranolol)
- Hyperthyroidism
 - Propranolol: Reduce symptoms of sympathetic over activity caused by the excess thyroxin
- Portal hypertension (Nadolol or propranolol)
- Glaucoma (topical use)
 - **Timolol**, levobunolol (nonselective BB), betaxolol (β_1 blocker)
 - Decrease aqueous humor production (blockage of B2 receptor)
 - They do not affect accommodation and pupillary reactions

Adverse effects of β -blockers

☛ CVS

- Bradycardia
- hypotension
- AV block

☛ Bronchoconstriction

☛ Hypoglycemic effect ($\downarrow$ glycogenolysis)

☛ Affect lipid profile ($\downarrow$ HDL, $\uparrow$ LDL&TG)

☛ Muscle pain & fatigue

☛ Sleep disturbances, nightmares

☛ Impaired sexual activity

☛ Reduce peripheral blood flow $\Rightarrow$ cold extremities

Produced by non-selective blockers

Contraindication

- Hypersensitivity reaction
- COPD & asthma (nonselective BBs cause bronchoconstriction)
- Hypotension
- Hypothyroidism
- Bradycardia
- Acute decompensated heart failure
- Combination with Ca-channel blockers
- Some general anesthetics-cardiac depression

Thank you!